ever, of one of our H-13 class engines, to which a full installation of flexible stay-bolts was applied three and a half years ago. Ten or twelve bolts in different parts of the fire-box were removed to discover whether there were any fractures or broken bolts. No defect of any kind was found in the bolts which were removed, and, although the writer personally inspected and sounded each bolt in the boiler, he could not discover one that was broken. He also applied straight edges along the sheet at different parts to see if there were any contortions or bulges which would denote a broken bolt. The fire-box, however, proved to be in normal condition, and this installation seems to prove the value of flexible stay-bolts and the high efficiency of the method in which they were applied.

B. F. Sarver.—The road I am connected with has applied 15 installations of flexible staybolts. These bolts have been examined from time to time. And we have never found any broken or defective bolts in any way. I personally think the flexible staybolt is doing as much good in preserving our fireboxes as anything we could possibly use.

The President (addressing Mr. Smythe).—At the American Locomotive Works you have removed a firebox that had flexible bolts?

J. H. Smythe.—We removed two boxes for the Chicago & Alton and I failed to discover one that had the least bit of a fracture. Two or three years ago one of our friends got up and said they found two. But to my mind there is nothing better for a boiler than flexible staybolts. If I was going to design a boiler I would have a total installation of flexible staybolts. I would have all the sides, and about three or four rows of sling-stays. I think that would eliminate your trouble with the top of the flue sheet cracking.

It is our standard practice to apply flexible staybolts in all high-pressure engines. We put 450 to 500 of these bolts in. We have removed some of the fireboxes and found some of these bolts, when we have been taking the box out, with the heads broken off. Some have been fractured. The location of these bolts have been in the outer row. We found about eight bolts on the outside row with the heads off, and eight or nine with the heads half broken. I believe these bolts are beneficial to the side sheets. We have no trouble with our side sheets.

We want to find a better method of testing these bolts. We have tried water pressure and steam pressure and air pressure. After these tests have been made we strip the jacket and the bolts we found broken we take out. We find better results with water pressure of 100 to 125 pounds. That is the way we are trying to locate the broken bolts.

Mr. McKeown.—We have been applying flexible staybolts to our throat sheets for the last five years, and I have yet to find the first broken flexible bolt. I think the bolt is all right

J. H. Smythe.—There are two things that could be the cause of broken staybolts; first, poor material; second, someone might put a greater strain on those bolts, and that might cause breakage. It has been suggested that we ease up those bolts. To my mind I think it is going to prove all right. If we put a greater strain on one bolt than another, you have got more presure on that bolt; and we cannot condemn the method because a few of them break.

It was voted that the report of the committee be accepted, and that the flexible bolt be recommended as the best known remedy for relieving the expansion and contraction of the boiler, and that the committee be discharged.

Best Method of Applying Flues, Best Method for Caring for Same while Engine is on the Road and at Terminals and Best Tools for Same.

The best method of applying flues is to have the flues fit perfectly tight in the copper. The copper should be tightened in the sheet with a sectional expander not rolled. Then tighten the flues with a roller expander and roll the lower edges tight enough to hold the flues while belling out. Then prosser them with a prossering tool 1-16 inch wider than the thickness of the sheet. After they are prossered bead them with a standard beading tool.

The best method for caring for flues while on the road is, in our opinion, to have the flues cleaned of all cinders and soot, to keep the temperature of steam as regular as possible, and to keep a clean fire. On the New York Central lines, particularly on the Lake Shore, we use arch tubes and a brick arch. This arch is a great protection for the flues, and the way the arch is applied right close to the flue sheet excludes the cold air from striking the flues. We have very little trouble with leaking flues, but as a precautionary measure at every washout, which occurs every seven days, we blow out the flues and clean off the top of the brick of all cinders and accumulation, if any. Also as a precautionary measure we give our flues a monthly prossering, whether they leak or not.

The best tools, in our opinion, are the prosser expander, and a standard beading tool to conform and take in 3-16 inch stock for the fire-box end. John German, Chairman.

C. L. Wilson.—On the Rock Island railroad we properly prepare the hole; and if the flue sheet has any superfluous metal on it we work it off. We cut it off with a milling tool. It does not make much to trim off. The flue holes are reamed out and the coppers are properly fitted and rolled. The flue is entered and pounded up. The flue is then beaded down two-thirds of the way, and the flue is prossered and calked. When the old sheet comes in it is wavey more or less. By rolling the flue and beading it down, we get the bead on the angle with the sheet. When you prosser the flue out, it works better. In beading the flue you disturb the flue in the sheet. You come along and take them all out by prossering the flue. Then when you calk your flues you are not going to disturb your flue in the hole.

We prosser by a long stroke hammer. We do not advocate the calking of flues with water or steam in the boiler. Under ordinary practice the flues are expanded, and we do not calk them. But when the engine comes in for washing we calk the flues.

A. N. Lucas.—We turn our flues over by hand; we bell them out and hit them with the flat of the hammer. We do all of our expanding with a 3½-pound maul, with a 20-inch handle.

J. W. Kelly.—We are about to adopt a time limit to leave flues in the boilers. We think three years a sufficient time for a flue to stay in a boiler.

W. H. Laughridge.—The Hocking Valley has adopted a rule to remove the flues after an engine had made so many miles, regardless of the condition of the flues. The time set was taken from the average miles run on the different divisions. We have found it to be a very good practice. Speaking of the setting of flues it seems to me that belling flues over with the hammer and unnecessary rolling is injurious to the flues.We drive them in, and the man uses a hammer to lip the flues so that they won't back out. After he gets the flues seated his expander rotates around until the flues are solid. We beat them down solid. We give them a very light rolling afterwards. I do not approve of working a flue any more than is absolutely necessary.

C. W. Lewis.—We do not use the hammer at all on flues. We never allow a man to put a hammer on flues. When we get the flues all in we begin to set them. We do not affect the end of the flues at all. Then we prosser them all over. Then we bead them. The less work you can do the better it is. I believe that boiler makers have to meet the conditions

THE INSTALLATION OF FLEXIBLE STAYBOLTS.

BY M. M. MCALLISTER.

In the application of Tate flexible staybolts to new fire boxes at the Collinwood Shops of the Lake Shore and Michigan Southern Railroad, we first drill out the old bolts with a twist drill ⅛ of an inch below the diameter of the bolt, having prick punched the bolt in the center, after chiselling off the end, and run the twist drill in a sufficient depth to pass the thickness of the sheets (about ½ inch) and then use a Wagstaff staybolt cutter to cut through the staybolt, leaving a thin shell of the old bolt in the sheet. When all the bolts have been drilled in this manner the sheets with the rivets removed are pulled out, and the shells of the old bolts in the outer sheet are collapsed with a chisel until separated, leaving the outer shell all ready for either a new sheet or a new fire-box, whichever the case might be.

In applying new fire-box sheets, after same have been located and riveted up to the mud-ring, the staybolt holes are drilled and tapped out with a regular drill and staybolt tap, run through both the inner and outer sheets, as ordinarily done; then we enlarge the holes in the outer sheet for the sleeves of the flexible stay-bolt, with a flat drill, made of high speed steel, running 280 revolutions per minute, and applied to a motor, following with a roughing taper reamer, which is well lubricated, which leaves the hole of the proper dimensions and taper for the tap which is used for the flexible staybolt sleeve.

In tapping the large holes for the flexible staybolt sleeves we first run a straight tap and then a taper tap, by motor whose power is compounded, at a speed of 280 revolutions per minute, using a collar on the taper tap as a gage to keep it from going in too far and tapping the holes too large for the required sleeve, to be used according to the thickness of the plate. The straight tap is used to block out the metal, and the taper tap is used to size up the hole, for the regular taper with full threads, using lubricant sufficient to keep the taps cool and make a nice smooth thread in the sheets.

In case new sheets are used for the outside, and also new side sheets for the fire-box, we drill holes ⅞ inch in diameter for 1 inch bolts, tapping the size for the bolt; then after tapping through both sheets with a regular tap, giving us a small hole in the inner sheets with a regular standard, the hole in the outer sheet is then drilled and tapped as before described. We believe this is the most practical way of cutting the holes in line, using, however, guide bars in the roughing reamer, straight tap and taper tap in machining the large hole in the outer sheet for the flexible staybolt sleeve.

It is a good practice to have two sets of tools, and while using one set allow the other to remain in oil to cool off, providing small buckets with sufficient oil to cover two-thirds of the tools, which are placed in the oil in upright positions.

In the use of the roughing reamer, straight tap and taper tap, guide bars should always be attached to the tools to engage with the hole in the inner sheet, affording a stiff connection for maintaining alignment.

Braces are bolted to each end of the sheet in the shape of forged arms, extending out as to take a three-inch plank, which extends from one arm to the other, known as "Old Men," used to brace and support the motor during the operations of machining and allowing for the use of two gangs of operators on the same side of the boiler at the same time, and by moving the braces and plank and following up the side of the boiler, the whole arrangement affords a ready means of making the operation of drilling comparatively easy and saves time.

The question has often been asked, "Why not punch the large holes?" In punching a hole for the flexible staybolt sleeve in half-inch or nine-sixteenths material, it is very apt to distort the material to such an extent that it is considered dangerous, and this practice should be discouraged, inasmuch as in the process of bending the outer sheet to conform to the contour lines of the boiler, the large holes that have been previously punched render the operation of bending unsatisfactory, and after the sheet is bent the punched holes do not ream up as perfectly as the enlarged holes before mentioned.

After the holes in the outer and inner sheets are tapped the flexible staybolt sleeves are screwed into the outer sheets by a stud nut, using a wrench about three feet long, until the sleeve is screwed to a steam-tight fit, after which we insert the bolt through the sleeve, screwing it into the threads of the inside sheet by using a bolt driver in a motor until the head of the bolt almost engages its seat, then screwing the bolt with a hand driver until it seats snugly, when it is turned back about one-quarter turn, as it is necessary to maintain a natural hanging of the sheets when all bolts are riveted up, and the riveting up has a slight tendency to draw the bolt up to the sleeve seat. This method is more apt to give an even tension over the entire area of sheet and is recommended as good practice.

To prevent sleeves from loosening in the outer sheet while the operation of riveting up the staybolt end which enters the fire-box is performed, we use instead of the ordinary dolly-bar and holder-on a bushing which screws over the sleeve of the flexible bolt, in the center of which is a plunger or pin with the end formed half round to conform to the shape of the bolt head and so contained that the pin remains at all times central to the staybolt during the riveting, while the jar or thrust from the dolly-bar held on the end is less liable to affect the sleeve. This arrangement, or something similar, is much better than holding-on in the usual way; in fact, it is necessary for the proper application of the flexible staybolt.

After the bolts have all been riveted up, it is well to gage each bolt head from the end of the sleeve where the cap makes its seat, to prove that the bolt is well seated in the sleeve and clearance is allowed between bolt head and cap. A good gage is made by sawing a cap in half and removing the threads, thus giving two gages which can be readily placed on the sleeve seat, showing the clearance left between the bolt head and cap.

With the foregoing operations properly performed and inspection carried on after each operation to insure that all machining and riveting is correct, the caps are then screwed on to the sleeve by first brushing the cap threads lightly with graphite and cylinder oil, mixed thin. The caps are screwed up tight, and should screw over the sleeve easily, so as to make a snug steam-tight fit on the face of the sleeve.

The sleeve face is squared up to engage with the similar surface in the cap, and before screwing on the cap it is well to inspect the sleeve surface on the end to see that no hammer marks or burrs will prevent the making of a tight joint. In case such are discovered they should be removed and no difficulty as to fit will be found.

Our experience proves the fact that when all machine operations are performed with care as to size, and general inspection made quickly following each operation, when all are found to be right, the assemblage when finished is perfect and little or no annoyance or difficulty will be found in the perfect working of the flexible stay.

The total cost of applying flexible staybolts, such as we apply, should not exceed 15 cents for labor, with one boiler maker at 32½ cents, and a helper at 18 cents per hour. When driving bolts one more boiler maker is required.

The new plant of the Western Dry Dock & Shipbuilding Company, at Port Arthur, Ont., is to include a large, completely equipped boiler shop.